Linux Syntax Cheat Sheet

1. File & Directory Navigation

Show current directory	`pwd`
List files	`ls`
List files with details and hidden files	`ls -lah /home/user/project`
List files, newest first	`ls -lt models/`
Go to a directory	`cd /mnt/data/datasets`
Go up one level	`cd ..`
Go to home directory	`cd ~/mlops-project`
Go back to previous directory	`cd -`
Find files by name	`find . -name "*.pkl"`
Find files by size	`find /data -type f -size +500M`
Find files modified recently	`find . -mtime -1`
Show location of a command	`which python3`
Show command, man page and source location	`whereis pip`

2. File & Directory Management

Create a directory	`mkdir models`
Create nested directories	`mkdir -p experiments/run_01/checkpoints`
Create an empty file	`touch requirements.txt`
Copy a file	`cp config.yaml config_backup.yaml`
Copy a directory	`cp -r dataset/ /mnt/backup/dataset/`
Move a file	`mv results.csv reports/results_2026.csv`
Rename a file	`mv model_v1.pt model_v2.pt`
Delete a file	`rm old_log.txt`
Delete a directory with contents	`rm -r old_folder/`
Force delete without prompts	`rm -rf tmp_checkpoints/`

3. Viewing and Editing File Contents

Display a file	`cat config.yaml`
Combine files into one	`cat file1.txt file2.txt > combined.txt`
View a file page by page	`less training.log`
Show first N lines	`head -n 20 data.csv`
Show last N lines	`tail -n 50 training.log`
Follow a file live	`tail -f training.log`
Edit with nano	`nano train_config.yaml`
Edit with vim	`vim Dockerfile`
Search text in a file	`grep "ERROR" training.log`
Search ignoring case	`grep -i "warning" app.log`
Search a folder with line numbers	`grep -rn "learning_rate" ./src`

4. File Permissions & Ownership

View permissions	`ls -l config.yaml`
Add read permission	`chmod +r notes.txt`
Add write permission	`chmod +w config.yaml`
Add execute permission	`chmod +x run_training.sh`
Add permission for owner only	`chmod u+x run_training.sh`
Add permission for group	`chmod g+w config.yaml`
Remove permission from others	`chmod o-w config.yaml`
Set exact permissions	`chmod u=rw,g=r,o=r notes.txt`
Set permissions with numbers (644 = rw-r--r--)	`chmod 644 config.yaml`
Set permissions with numbers (755 = rwxr-xr-x)	`chmod 755 run_training.sh`
Change permissions of a folder and its contents	`chmod -R 755 scripts/`
Change file owner	`sudo chown user model.pt`
Change owner and group	`sudo chown user:user model.pt`
Change owner of a folder and its contents	`sudo chown -R user:user ./outputs`
Run a command as administrator	`sudo apt update`

5. Process & Job Management

List all running processes	`ps aux`
Find a specific process	`ps aux | grep python`
Live process monitor	`top`
Better live process monitor	`htop`
Stop a process gracefully	`kill 12345`
Force stop a process	`kill -9 12345`
Run a command in the background	`python train.py &`
Run in background and keep alive after logout	`nohup python train.py > train.log 2>&1 &`
List background jobs	`jobs`
Bring a job to the foreground	`fg %1`
Resume a job in the background	`bg %1`

6. System Monitoring & Storage

Show disk space	`df -h`
Show size of a folder	`du -sh datasets/`
Show folder sizes sorted	`du -h --max-depth=1 . | sort -h`
Show memory usage	`free -h`
Show GPU status	`nvidia-smi`
Refresh GPU status every N seconds	`watch -n 1 nvidia-smi`